

The integral monodromy of the (3,4,∞) period system of the S^6 construction: an audited record with new results

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Abstract

The candidate complex structure on S^6 circulated by Levent Alpöge is built from a family of complex 2-tori over the projective line whose integral period system is a pair of matrices $T_1, T_2 \in GL_4(\mathbb{Z})$ with $T_1^3 = T_2^4 = I$, preserving an alternating form Q_0 of type (1,6) and fixing a primitive isotropic vector γ . We determine the monodromy group $G = \langle T_1, T_2 \rangle$ exactly: it is the kernel of an explicit character Φ of order 12 on the stabilizer $G_{\mathbb{Z}} = SL_2(\mathbb{Z}) \ltimes \text{Heis}(\mathbb{Z})$ of γ , a normal congruence subgroup of exact level 12 with cyclic quotient, arithmetic in its Zariski closure. The Levi half of Φ is precisely the marked peripheral character of the author's higher-torus update, and its Heisenberg half is the multiplier of the odd Jacobi theta function, so the update's character is the invariant that cuts the monodromy group out of its arithmetic hull. We also classify, at every level, the orbits of the dual action that the Erdős–Straus workbench uses to build its (3,4,∞) covers.

0. Introduction

0.1 Why this record

The S^6 construction [A] was examined in a validate/repair/disprove record produced with AI systems [R], which the author extended with a higher-torus and cusp update (files 14_ and 15_ of [R]). The integral period system is the part of the construction that can be determined completely by algebra, and it is used elsewhere: the author's higher-torus update attaches a character to its peripheral monodromy, and the Erdős–Straus workbench builds covers of the projective line from its dual action. This record determines the monodromy group exactly and explains how these two uses fit it.

0.2 Main results

- **Theorem 3.1.** $G = \ker \Phi$ for an explicit surjective character $\Phi: G_{\mathbb{Z}} \rightarrow \mathbb{Z}/12$. So G is normal of index 12 in $G_{\mathbb{Z}}$, with cyclic quotient generated by the central transvection; it is a congruence subgroup of exact level 12; its Levi image is all of $SL_2(\mathbb{Z})$, while its intersection with the Levi subgroup is the commutator subgroup; and it is Zariski dense and arithmetic in the algebraic stabilizer of γ .
- **Propositions 4.1–4.2.** The Heisenberg part of G is the subgroup cut out by the odd quadratic refinement of the mod-2 form, equivalently the kernel of the Heisenberg multiplier of the odd Jacobi theta function; G is one of twelve twists of it by the abelianization of $SL_2(\mathbb{Z})$, and the extension does not split.
- **Theorem 5.1.** The marked peripheral character χ of the author's update equals $-ab$, its values $(-1, 4, -3)$ are those of ab on the Levi parts of T_0, T_1, T_2 , and on G it is determined by the Heisenberg coordinates: $\chi(M) \equiv s - 6(p^2 + pr + r^2) \pmod{12}$.
- **Theorem 6.1.** At every level D the orbits of G on the affine hyperplane H_D are classified by $\gcd(b, c, \gcd(D, 6))$, and they coincide with the orbits of $G_{\mathbb{Z}}$.

0.3 What the results mean

The monodromy group is as large as it can be subject to one constraint, and the constraint is the author's peripheral character: G is exactly the locus where the Levi character and the Heisenberg theta multiplier cancel. So the update's character is not an auxiliary label but the invariant that separates the monodromy from its arithmetic hull. Because G is arithmetic rather than thin, the tools of congruence subgroups apply, and they give the complete orbit classification that the Erdős–Straus covers need. The Heisenberg part links the period system to spin structures and theta characteristics, and the ambient group $Sp(Q_0, \mathbb{Z})$ is

the paramodular group of level 6 of Gritsenko and Hulek, which suggests the identification of Φ with one of their characters (Question 1).

0.4 How this record was made

This is an audit, by an AI system, of one part of the S^6 material: the integral period system and its monodromy group. The statements used from [A] and [R] were re-derived from their sources, all statements were checked by computer, and the results of §§3–6 are new. The existence question for the complex structure on S^6 itself is outside the scope of this record and is not verified here. Statements carry one of four statuses, as in the companion papers: *verified* (proved here or re-computed by a program listed in §9), *not verified here*, *my assessment* (a view with its reasons, in the first person), and *proved negative* (with the counterexample or derivation in the text). This record replaces §§1–3 of note 52_ and the corresponding part of the S^6 reader, which were withdrawn on 27 September 2026.

0.5 Organization

Part A (§§1–6) contains the verified results: the setting (§1), the stabilizer of γ (§2), the main theorem (§3), the Heisenberg part (§4), the peripheral character (§5) and the orbits (§6). Part B (§7) contains the proved negative statements, Part C (§8) the connections to other areas with my assessment, §9 the verification and §10 open questions.

Part A. Verified results

1. Setting

The period system is taken from the manuscript [A], as reconstructed in file 03_ of the record [R], §2. Let $V = \mathbb{Z}^4$ with basis (γ, u, w, δ) . The local monodromies at the two elliptic points are, by formula (2.2) of [A],

- $T_1: \gamma \mapsto \gamma, u \mapsto -u - w, w \mapsto -6\gamma + u, \delta \mapsto 2\gamma + u + w + \delta,$
- $T_2: \gamma \mapsto \gamma, u \mapsto 6\gamma + w, w \mapsto -u, \delta \mapsto -3\gamma + u + \delta,$

and the monodromy at the cusp is $T_0 = (T_1 T_2)^{-1}$. By Lemma 2.2 of [A], $T_1^3 = T_2^4 = I$, and $T_0 = I + N$ with $N^2 = 0$, $N\gamma = Nu = 0$, $Nw = -u$, $N\delta = \gamma$, so $\ker N = \text{im } N = \langle \gamma, u \rangle$.

By Lemma 2.8 of [A], the group of $\langle T_1, T_2 \rangle$ -invariant alternating forms is infinite cyclic, generated by Q_0 with $Q_0(\gamma, \delta) = 1$ and $Q_0(u, w) = 6$, all other pairings of basis vectors being zero. Its elementary divisors are $(1, 1, 6, 6)$, so Q_0 has type $(1, 6)$. We write $G = \langle T_1, T_2 \rangle$ and

$$G_{\mathbb{Z}} = \{ g \in \text{Sp}(Q_0, \mathbb{Z}) : g\gamma = \gamma \}.$$

Every T_i fixes γ and preserves Q_0 , so $G \subseteq G_{\mathbb{Z}}$.

2. The stabilizer of γ

For $M \in \text{SL}_2(\mathbb{Z})$, let L_M act by M on the plane $\langle u, w \rangle$, in the basis (u, w) , and fix γ and δ . For $p, r, s \in \mathbb{Z}$, let $h(p, r, s)$ be the map

$$\gamma \mapsto \gamma, u \mapsto u - 6r\gamma, w \mapsto w + 6p\gamma, \delta \mapsto \delta + pu + rw + s\gamma.$$

Lemma 2.1. Every element of $G_{\mathbb{Z}}$ can be written uniquely as $L_M h(p, r, s)$. The elements $h(p, r, s)$ form a group $\text{Heis}(\mathbb{Z})$ with the law

$$h(p_1, r_1, s_1) h(p_2, r_2, s_2) = h(p_1 + p_2, r_1 + r_2, s_1 + s_2 + 6(p_1 r_2 - p_2 r_1)),$$

and $L_M h(p, r, s) L_M^{-1} = h(M(p, r), s)$. Hence $G_{\mathbb{Z}} = \text{SL}_2(\mathbb{Z}) \ltimes \text{Heis}(\mathbb{Z})$.

Proof. Let $g \in G_{\mathbb{Z}}$. Since g fixes γ , it preserves $\gamma^\perp = \langle \gamma, u, w \rangle$. The form induced on $\gamma^\perp / \langle \gamma \rangle$ is $6 \cdot (u^* \wedge w^*)$, so the induced map M has determinant 1. On $V/\gamma^\perp$, g acts trivially, since $Q_0(\gamma, gx) = Q_0(g^{-1}\gamma, x) = Q_0(\gamma, x)$.

Then $L_M^{-1}g$ acts trivially on $\langle \gamma \rangle$, on $\gamma^\perp / \langle \gamma \rangle$ and on $V/\gamma^\perp$. Such a unipotent map has the form $u \mapsto u + a_u \gamma$, $w \mapsto w + a_w \gamma$, $\delta \mapsto \delta + pu + rw + sy$. It preserves Q_0 exactly when $a_u = -6r$ and $a_w = 6p$. The law and the conjugation formula are matrix products. ■

Two consequences are used repeatedly. Inside $\text{Heis}(\mathbb{Z})$, conjugation acts by $h(p_0, r_0, s_0) h(p, r, s) h(p_0, r_0, s_0)^{-1} = h(p, r, s + 12(p_0 r - p r_0))$. The centre of $G_{\mathbb{Z}}$ is $\{h(0, 0, s)\} = \langle t_\gamma \rangle$, where $t_\gamma = h(0, 0, 1)$ is the transvection $\delta \mapsto \delta + \gamma$, that is, $x \mapsto x + Q_0(\gamma, x)\gamma$.

The subgroup $H(\mathbb{Z}) = \{h(p, r, 6\kappa)\}$ carries the standard Heisenberg law $\kappa_1 + \kappa_2 + (p_1 r_2 - p_2 r_1)$. It has index 6 in $\text{Heis}(\mathbb{Z})$. So $G_{\mathbb{Z}}$ contains the Jacobi group $\text{SL}_2(\mathbb{Z}) \ltimes H(\mathbb{Z})$ with index 6, and is generated by it together with t_γ .

3. The monodromy group is the kernel of a character of order 12

Let $\text{ab}: \text{SL}_2(\mathbb{Z}) \rightarrow \mathbb{Z}/12$ be the abelianization, normalized by $\text{ab}(T) = 1$ for $T = [[1, 1], [0, 1]]$. Then $\text{ab}(S) = 9$ for $S = [[0, -1], [1, 0]]$, $\text{ab}(R) = 4$ for $R = [[-1, 1], [-1, 0]]$, and $\text{ab}(-I) = 6$. These values are forced:

- $\text{SL}_2(\mathbb{Z})^\wedge \text{ab} \cong \mathbb{Z}/12$ is generated by the class of T ; this is standard, and $[R, 15]$ proves it by a Smith normal form in Theorem 6.1;
- S and R have orders 4 and 3;
- $\text{ab}(S) + \text{ab}(R) = \text{ab}(SR) = \text{ab}(T)$, because $SR = [[1, 0], [-1, 1]]$ is conjugate to T .

On $\text{Heis}(\mathbb{Z})$ put

$$\varphi(h(p, r, s)) = s - 6q(p, r) \pmod{12}, \text{ with } q(p, r) = p^2 + pr + r^2 \pmod{2},$$

and on $G_{\mathbb{Z}}$ put $\Phi(L_M h) = \text{ab}(M) + \varphi(h) \pmod{12}$.

Theorem 3.1.

1. $\Phi: G_{\mathbb{Z}} \rightarrow \mathbb{Z}/12$ is a surjective homomorphism, and $G = \ker \Phi$.
2. The local monodromies decompose as $T_1 = L_R h(-1, 0, 2)$, $T_2 = L_S h(0, -1, -3)$ and $T_0 = L_{\{T^{-1}\}} t_\gamma$. In particular the Levi image of G is all of $\text{SL}_2(\mathbb{Z})$.
3. G is normal in $G_{\mathbb{Z}}$, and $G_{\mathbb{Z}}/G \cong \mathbb{Z}/12$ is cyclic, generated by the class of the central transvection t_γ . Moreover $G_{\mathbb{Z}} = G \cdot \langle t_\gamma \rangle$ and $G \cap \langle t_\gamma \rangle = \langle t_\gamma^{12} \rangle$.
4. $G \cap \text{Heis}(\mathbb{Z}) = H' := \{h(p, r, s) : s \equiv 6(p^2 + pr + r^2) \pmod{12}\}$. This subgroup has index 12, with coset representatives $h(0, 0, k)$, $0 \leq k \leq 11$. It is generated by the two elements $c = [T_1, T_2^2] = h(1, 0, -6)$ and $c' = T_2 c T_2^{-1} = h(0, 1, 6)$.
5. $G \cap \text{SL}_2(\mathbb{Z}) = \text{SL}_2(\mathbb{Z})'$, the commutator subgroup, which is free of rank 2 and of index 12.
6. The abelianization of $G_{\mathbb{Z}}$ is $(\mathbb{Z}/12)^2$, via $g \mapsto (\text{ab}(M), \varphi(h))$, and $[G_{\mathbb{Z}}, G_{\mathbb{Z}}] = \text{SL}_2(\mathbb{Z})' \ltimes H'$.
7. G contains the principal congruence subgroup of level 12, and no principal congruence subgroup of level 4 or 6. For every $D \geq 1$, the image of G in $\text{GL}_4(\mathbb{Z}/D)$ has index $\gcd(D, 12)$ in the image of $G_{\mathbb{Z}}$.
8. G is Zariski dense in $P = \text{SL}_2 \ltimes \text{Heis}$, the algebraic stabilizer of γ , and it is an arithmetic subgroup of P , since it has finite index in $P(\mathbb{Z}) = G_{\mathbb{Z}}$.

Proof.

(1), Φ is a homomorphism. φ is a homomorphism on $\text{Heis}(\mathbb{Z})$, because q is a quadratic refinement of the mod-2 symplectic form $\omega(x, y) = p_x r_y - p_y r_x$:

$$q(x + y) \equiv q(x) + q(y) + \omega(x, y) \pmod{2},$$

and therefore $6q(x + y) \equiv 6q(x) + 6q(y) + 6\omega(x, y) \pmod{12}$. Moreover φ is invariant under the Levi action. q equals 1 exactly on the three nonzero vectors of $\mathbb{F}_2^2$, so $q(Mx) \equiv q(x)$. Writing $L_{\{M_1\}} h_1 \cdot L_{\{M_2\}} h_2 = L_{\{M_1 M_2\}} \cdot (L_{\{M_2\}}^{-1} h_1 L_{\{M_2\}}) \cdot h_2$ gives $\Phi(g_1 g_2) = \Phi(g_1) + \Phi(g_2)$.

(1), $G \subseteq \ker \Phi$ and the lower bound. By (2), $\Phi(T_1) = 4 + (2 - 6) = 0$ and $\Phi(T_2) = 9 + (-3 - 6) = 0$, so $G \subseteq \ker \Phi$. Since $\Phi(t_\gamma) = 1$, $\ker \Phi$ has index 12. Hence $[G_{\mathbb{Z}} : G] \geq 12$ and $G \cap \text{Heis}(\mathbb{Z}) \subseteq \ker \varphi = H'$.

(1), $G = \ker \Phi$. The words $c^\wedge p c' \wedge r [c, c']^\wedge m$ equal $h(p, r, -6p + 6r + 6pr + 12m)$. Since $-6p + 6r + 6pr \equiv 6q(p, r) \pmod{12}$, these words exhaust H' , so $H' = \langle c, c' \rangle \subseteq G$. Since R and S generate $\text{SL}_2(\mathbb{Z})$ ($S(SR)S^{-1} = T$), G maps onto $\text{SL}_2(\mathbb{Z})$. Therefore

$$[G_{\mathbb{Z}} : G] = [\text{Heis}(\mathbb{Z}) : G \cap \text{Heis}(\mathbb{Z})] \leq [\text{Heis}(\mathbb{Z}) : H'] = 12,$$

and $G = \ker \Phi$.

(2) is a direct computation.

(3) follows from (1) and from the description of the centre in §2.

(4) H' is $\ker \varphi$. The identity $h(0, 0, k) h(p, r, s) = h(p, r, s + k)$ gives the coset representatives.

(5) $L_M \in G$ exactly when $\text{ab}(M) = 0$. The commutator subgroup $\text{SL}_2(\mathbb{Z})'$ has index 12 and is torsion-free:

- every element of finite order in $\text{SL}_2(\mathbb{Z})$ is conjugate to a power of S or of $-R$;
- these powers have ab -values 9, 6, 3 and 10, 2, 8, 4, 6, all nonzero.

A torsion-free subgroup of the amalgam $\mathbb{Z}/4 *_{\{\mathbb{Z}/2\}} \mathbb{Z}/6$ acts freely on its Bass-Serre tree and is free [Se, Ch. I]. Its rank follows from the Euler characteristic: $12 \cdot (-1/12) = -1 = 1 - \text{rank}$.

(6) The map is a homomorphism by the computation in (1), applied to each coordinate. It is onto, since $L_T \mapsto (1, 0)$ and $t_\gamma \mapsto (0, 1)$. Its kernel $\text{SL}_2(\mathbb{Z})' \rtimes H'$ is generated by commutators:

- the commutators of Levi elements give $\text{SL}_2(\mathbb{Z})'$;
- the commutators $[L_M, h(x, s)] = h((M - I)x, -6\omega(Mx, x))$ have projections $(M - I)x$ that span $\mathbb{Z}^2$, for example $(T - I)(0, 1) = (1, 0)$ and $(T^T - I)(1, 0) = (0, 1)$;
- the commutators of these give $h(0, 0, 12)$.

By Proposition 4.1 the subgroup they generate is H' .

(7) If $g \equiv I \pmod{12}$, then $\text{ab}(M) = 0$, because ab factors through $\text{SL}_2(\mathbb{Z}/12)$ (checked by a breadth-first search over all 1152 elements, §9), and $\varphi(h) = 0$. On the other hand $h(0, 0, 4)$ and $h(0, 0, 6)$ lie in the level-4 and level-6 principal congruence subgroups and have $\Phi = 4$ and 6.

For the index, write K_D for the principal congruence subgroup of level D . Then $[G_{\mathbb{Z}} \bmod D : G \bmod D] = [G_{\mathbb{Z}} : G K_D] = 12/|\Phi(K_D)|$.

- Since $t_\gamma^D \in K_D$, $\Phi(K_D)$ contains $\gcd(D, 12)\mathbb{Z}/12$.
- Conversely, $\Phi \bmod d$ factors through reduction mod d for $d = \gcd(D, 12)$: $\text{SL}_2(\mathbb{Z}/12) = \text{SL}_2(\mathbb{Z}/4) \times \text{SL}_2(\mathbb{Z}/3)$ and $\text{SL}_2(\mathbb{Z}/3)^{\text{ab}} = \mathbb{Z}/3$, and $6q(p, r) \bmod d$ depends only on $p, r \bmod 2$.

(8) $h(2, 0, 0)$, $h(0, 2, 0)$ and $h(0, 0, 12)$ lie in $H' \subseteq G$. Their cyclic groups are Zariski dense in the three one-parameter subgroups that generate Heis , so the Zariski closure of G contains Heis . Its Levi image is closed and contains $\text{SL}_2(\mathbb{Z})$, which is dense in SL_2 . ■

In particular G is not a thin group in the sense of Sarnak [Sa]: it has finite index in the integral points of its Zariski closure. This places the S^6 period system among the arithmetic cases of rank-4 symplectic monodromy. Brav and Thomas [BT] exhibit hypergeometric monodromy groups in $\text{Sp}(4, \mathbb{Z})$ of the other, thin kind.

4. The Heisenberg part, and the twist

Proposition 4.1.

1. The quadratic refinements of $\omega \bmod 2$ on $\mathbb{F}_2^2$ are the four functions $\alpha p^2 + pr + \beta r^2$ ($\alpha, \beta \in \mathbb{F}_2$). Three have Arf invariant 0, and $q = p^2 + pr + r^2$ has Arf invariant 1. $\text{SL}_2(\mathbb{F}_2)$ permutes the three even refinements and fixes q .
2. Let $N \subseteq \text{Heis}(\mathbb{Z})$ be a subgroup that is normalized by the Levi group and maps onto $\mathbb{Z}^2$ under (p, r) . Then $N = N_k := \{h(p, r, s) : s \equiv 6q(p, r) \bmod k\}$ for a unique divisor k of 12. Conversely, every N_k is such a subgroup, of index k .
3. $H' = N_{12}$ is the smallest of them, and $H' = [G_{\mathbb{Z}}, \text{Heis}(\mathbb{Z})] = [G_{\mathbb{Z}}, G_{\mathbb{Z}}] \cap \text{Heis}(\mathbb{Z})$.
4. On $H(\mathbb{Z}) = \{h(p, r, 6\kappa)\}$, H' is the kernel of the character $v_H(h(p, r, 6\kappa)) = (-1)^{\{p + r + pr + \kappa\}}$.

Proof. (1) is a finite check.

- (2) Lifts of $(1, 0)$ and $(0, 1)$ in N have commutator $h(0, 0, 12)$. So N meets the centre in $k\mathbb{Z}$ for some k dividing 12, and $N_x = \sigma(x) + k\mathbb{Z}$ for each $x \in \mathbb{Z}^2$. The function $\lambda(x) = \sigma(x) - 6q(x) \pmod{k}$ is additive, by the law and (1), and it is Levi-invariant. It therefore vanishes on every $(M - I)\mathbb{Z}^2$, and these span $\mathbb{Z}^2$. So $\lambda = 0$.
- (3) The subgroup $[G_{\mathbb{Z}}, \text{Heis}(\mathbb{Z})]$ is Levi-normal, maps onto $\mathbb{Z}^2$, contains $h(0, 0, 12)$ and lies in H' . So it is $N_{12} = H'$.
- (4) With $s = 6\kappa$, the condition $s \equiv 6q(p, r) \pmod{12}$ says $\kappa \equiv q(p, r) \equiv p + pr + r \pmod{2}$. ■

Two readings of Proposition 4.1 connect the Heisenberg part to established theory.

Quadratic refinements of the mod-2 intersection form of a surface are its spin structures, with the Arf invariant as their parity [Jo]. On a Riemann surface they are its theta characteristics [At]. On a torus, the odd spin structure is the one fixed by the mapping class group $SL_2(\mathbb{Z})$, and (2) shows that SL_2 -invariance forces it.

The character v_H of (4) is the Heisenberg multiplier of the odd Jacobi theta function ϑ . Gritsenko and Hulek [GH, (1.4)–(1.5)] show that $\vartheta \in J_{\{1/2, 1/2\}}(v_{\eta^3} \times v_H)$. So the Heisenberg part of the monodromy group is the kernel of the odd theta multiplier.

Proposition 4.2. For $k \in \mathbb{Z}/12$ put $G_k = \{L_M h : \varphi(h) + k \cdot \text{ab}(M) \equiv 0\}$. The G_k are twelve distinct normal subgroups of index 12. Each has Levi image $SL_2(\mathbb{Z})$ and Heisenberg part H' , and $G = G_1$. The extension $1 \rightarrow H' \rightarrow G \rightarrow SL_2(\mathbb{Z}) \rightarrow 1$ does not split, whereas $G_0 = SL_2(\mathbb{Z}) \ltimes H'$ does.

Proof. Distinctness: $L_T h(0, 0, s) \in G_k$ exactly when $s \equiv -k$.

Non-splitting: a splitting σ would provide an involution $\sigma(-I) \in G$ over $-I$, commuting with an element $\sigma(R) \in G$ over R .

- Since $(L_{-I} h(x, s))^2 = h(0, 2s)$, the involutions of G over $-I$ are the $L_{-I} h(x, 0)$ with $\Phi = 6 - 6q(x) \equiv 0$, that is, with $x \not\equiv 0 \pmod{2}$.
- Conjugation by $L_R h(y, t)$ sends $L_{-I} h(x, 0)$ to $L_{-I} h(R(x - 2y), 0)$. Commuting would therefore require $Rx \equiv x \pmod{2}$.
- But R has no nonzero fixed vector on $\mathbb{F}_2^2$. ■

So the pair “index 12 in the centre, odd refinement mod 2” determines the Heisenberg part H' . The group G itself is the twist of H' by the abelianization of the Levi part.

5. The marked peripheral character of the S^6 update

The higher-torus update of the S^6 record [R, 15_, §3.2 and Theorem 6.1; see also 14_, §2] works with the peripheral matrices

$$C_1 = [[-1, -1], [1, 0]], C_2' = [[0, 1], [-1, 0]], C_0 = [[1, 1], [0, 1]].$$

It defines the marked peripheral character $\chi = \nu \circ \text{ab}: SL_2(\mathbb{Z}) \rightarrow \mathbb{Z}/12$ by $\chi(C_0) = -1$, and shows that $(\chi(C_0), \chi(C_1), \chi(C_2')) = (-1, 4, -3)$.

Theorem 5.1.

- $C_1 = \pi(T_1)^T$, $C_2' = \pi(T_2)^T$ and $C_0 = T$, where π denotes the Levi part, and $\chi = -\text{ab}$.
- The triple $(-1, 4, -3)$ equals $(\text{ab } \pi(T_0), \text{ab } \pi(T_1), \text{ab } \pi(T_2))$.
- For every $g = L_M h(p, r, s)$ in the monodromy group G ,

$$\chi(M) \equiv s - 6(p^2 + pr + r^2) \pmod{12}.$$

Proof. The matrix identities are direct. χ and $-\text{ab}$ are homomorphisms that agree on T , which generates the abelianization, so they are equal. The triple follows from Theorem 3.1(2). The last formula is $\Phi(g) = 0$ rewritten. ■

So the character that the update attaches to the peripheral monodromy is exactly the Levi half of the character whose kernel is the monodromy group. On G it is determined by the Heisenberg coordinates, and the monodromy group is the locus where the two halves cancel.

The same update also exhibits a Lie-framed torus whose spin quadratic form is $q(x, y) = x + xy + y$ over $\mathbb{F}_2$, with Arf invariant one [R, 15_, (147); 14_, §1]. Since $x + xy + y \equiv x^2 + xy + y^2 \pmod{2}$, this is the odd refinement of Proposition 4.1(1). Whether that torus's first homology is the lattice $\langle \tilde{u}, \tilde{w} \rangle$ of Proposition 4.1 is Question 2.

6. The orbits of the $(3, 4, \infty)$ covers at every level

The Erdős–Straus workbench (its package ES-09) uses the dual matrices $A_j = (T_j^{-1})^T$, acting on $\Lambda = \text{Hom}(V, \mathbb{Z})$ with dual basis $(\hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta})$, to build covers from the orbits of the affine hyperplane

$$H_D = \{\lambda \in \Lambda/D\Lambda : \hat{\gamma}\text{-coordinate of } \lambda = 1\}.$$

Write $\lambda = \hat{\gamma} + b\hat{u} + c\hat{w} + d\hat{\delta}$ and $e = \gcd(D, 6)$.

Theorem 6.1. For every $D \geq 1$, the orbits of G on H_D coincide with the orbits of $G_{\mathbb{Z}}$. The orbit of λ is $\{\lambda' \in H_D : \gcd(b', c', e) = \gcd(b, c, e)\}$,

of size $D^3 \cdot \#\{v \in (\mathbb{Z}/e)^2 : \gcd(v, e) = \gcd(b, c, e)\}/e^2$.

Proof.

The dual action. The dual action of $h(p, r, s)$ is $(b, c, d) \mapsto (b + 6r, c - 6p, d - s - pb - rc)$, and L_M acts on (b, c) by M^{-T} and fixes d .

The orbits of $G_{\mathbb{Z}}$. d can be moved freely, and (b, c) matters only modulo $6(\mathbb{Z}/D)^2$. So the $G_{\mathbb{Z}}$ -orbits correspond to the $\text{SL}_2(\mathbb{Z}/e)$ -orbits on $(\mathbb{Z}/e)^2$, which are classified by content, since $\text{SL}_2(\mathbb{Z})$ maps onto $\text{SL}_2(\mathbb{Z}/e)$.

G has the same orbits. G is normal and $G_{\mathbb{Z}} = G\langle t_{\gamma} \rangle$ with t_{γ} central, so it suffices to show $t_{\gamma}\lambda \in G\lambda$.

- At a point with $(b, c) = (g, 0)$, the Levi element $L_{\{(T^T)^{-1}\}}$ fixes λ and has $\Phi = 1$. Hence $t_{\gamma}L_{\{(T^T)^{-1}\}}^{-1} \in G$ moves λ exactly as t_{γ} does.
- Every point is $G_{\mathbb{Z}}$ -equivalent to one with $c = 0$.
- Normality and centrality transport the conclusion to every point. ■

Corollary 6.2.

- For $D = 2^k$ there are exactly two orbits, of sizes $D^3/4$ and $3D^3/4$, for every $k \geq 1$.
- For odd D there is one orbit if $3 \nmid D$, and there are two, of sizes $D^3/9$ and $8D^3/9$, if $3 \mid D$.
- For $6 \mid D$ there are four orbits. At $D = 6$ their sizes are 6, 18, 48 and 144; at $D = 12$ they are 48, 144, 384 and 1152.
- In general the orbits at $D = 2^k m$ (m odd) are the products of those at 2^k and at m .

This settles the question, raised when these covers were examined at even levels for $k \leq 5$, whether the two-orbit structure at $D = 2^k$ persists for all k and whether mixed levels decompose as products.

Part B. Proved negative statements

7. What the period system does not have

Three statements of Part A and of the manuscript are negative in form, and each is proved.

- **The extension does not split** (Proposition 4.2). No subgroup of G maps isomorphically onto $\text{SL}_2(\mathbb{Z})$: an involution over $-I$ would have to commute with an element over R , and R has no non-zero fixed vector on $\mathbb{F}_2^2$.

- **The level is exactly 12** (Theorem 3.1(7)). G contains no principal congruence subgroup of level 4 or 6, since $h(0, 0, 4)$ and $h(0, 0, 6)$ lie in them and have $\Phi = 4$ and 6.
- **No polarized limit mixed Hodge structure at the cusp** (the manuscript's Remark 3.23). With N as in §1, the form $b_N(x, y) = Q_0(x, Ny)$ on $V/\ker N = \langle w, \delta \rangle$ has $b_N(w, w) = Q_0(w, -u) = 6$, $b_N(\delta, \delta) = Q_0(\delta, y) = -1$ and $b_N(w, \delta) = b_N(\delta, w) = 0$. So $b_N = \text{diag}(6, -1)$ is indefinite, and (V, Q_0, N) does not support a polarized limit mixed Hodge structure.

These results say which standard tools apply as they stand. Methods that need a split extension or a polarized limit do not; methods for congruence subgroups of level 12 and for arithmetic groups do.

Part C. Connections to other areas

8. Further structure and connections

The period system connects to several established theories. For each connection this section states what is established and what is open, and gives my assessment.

The paramodular group of level 6. Gritsenko and Hulek [GH, §1] call $\text{Sp}(W_t, \mathbb{Z})$, for an alternating form W_t of type $(1, t)$, the integral paramodular group Γ_t . So $\text{Sp}(Q_0, \mathbb{Z}) \cong \Gamma_6$. Its maximal parabolic subgroups at one-dimensional cusps are the Jacobi groups extended by central elements. The parabolic that [GH] display fixes a vector of divisor t , corresponding to our u , while $G_{\mathbb{Z}}$ is the parabolic at the other cusp type, of divisor 1. It has the same shape: it contains $\text{SL}_2(\mathbb{Z}) \ltimes H(\mathbb{Z})$ with index 6 (§2). [GH, Lemma 1.1] also gives characters χ_3 and χ_4 of orders 3 and 4 of Γ_6 , whose restrictions to the Heisenberg group involve v_H ; Proposition 4.1(4) identifies v_H with the Heisenberg half of Φ . Whether Φ is the restriction of $\chi_3\chi_4^{-1}$ to $G_{\mathbb{Z}}$ is not verified here and is Question 1. My assessment: I think this identification is likely, because the two characters agree on the Heisenberg part and have the same order, but the relative normalization at the two cusp types has to be computed before it can be stated.

Spin structures and theta characteristics. Quadratic refinements of the mod-2 intersection form of a surface are its spin structures, with the Arf invariant as their parity [Jo]; on a Riemann surface they are its theta characteristics [At]. On a torus, the odd spin structure is the one fixed by the mapping class group $\text{SL}_2(\mathbb{Z})$, and Proposition 4.1(2) shows that SL_2 -invariance forces it in the monodromy group. The update exhibits a Lie-framed torus whose spin form $q(x, y) = x + xy + y$ has Arf invariant one [R, 15_, (147); 14_, §1]; since $x + xy + y \equiv x^2 + xy + y^2 \pmod{2}$, this is the odd refinement of Proposition 4.1(1). Whether that torus's first homology is the lattice $(\bar{u}, \bar{w})$ of Proposition 4.1 is not verified here (Question 2). My assessment: if it is, the update realizes geometrically the spin structure that cuts out the Heisenberg part of the monodromy, which would make the update's construction and the monodromy computation two descriptions of the same object.

The cusp. The cusp monodromy has $N^2 = 0$ and rank $N = 2$, with kernel and image the Lagrangian $\langle y, u \rangle$ [A, Lemma 2.2]. On $\Lambda^2 V$ the induced nilpotent N_2 satisfies $N_2^2 \neq 0$ and $N_2^3 = 0$. This is automatic for rank 2, since $N_2^2(a \wedge b) = 2Na \wedge Nb$, and it is the Jordan type of Kulikov's type III for H^2 of semistable degenerations of surfaces with trivial canonical class [Ku, PP]. The manuscript's central fibre at the cusp [A, §4] has the combinatorics of Mumford's toric degenerations of abelian surfaces: dP_6 with opposite sides of its hexagon identified, two triple points, and dual complex the A_2 -triangulation of the torus. Since b_N is indefinite (§7), the typed comparison with the Kulikov-Persson-Pinkham picture holds at the level of Jordan type and of combinatorics, and a Hodge-theoretic comparison needs a replacement for the polarization. My assessment: I think the combinatorial match is a strong indication that the degeneration is of Mumford type, and finding the right substitute for the polarization is the natural next step.

The modular part. Up to the factor 6 in the induced form, the Levi group acts on $\gamma^\perp / \langle \gamma \rangle$ as $\text{SL}_2(\mathbb{Z})$ acts on the first homology of an elliptic curve, and the Heisenberg part records the extension data. Theorem 3.1 and Proposition 4.1 describe both exactly. Whether $\gamma^\perp / \langle \gamma \rangle$ is the homology of an elliptic family inside or under the tori is Question 3.

Arithmetic, not thin. G has finite index in the integral points of its Zariski closure, so it is not thin in the

sense of Sarnak [Sa]. This places the S^6 period system among the arithmetic cases of rank-4 symplectic monodromy; Brav and Thomas [BT] exhibit hypergeometric monodromy groups in $\mathrm{Sp}(4, \mathbb{Z})$ of the thin kind. Arithmeticity makes strong approximation and the orbit classification of §6 available, and the Erdős–Straus workbench uses exactly this action to build its covers.

9. Verification

The statements were checked by three independent scripts, which are in the record’s checks/s6/ folder with their outputs. Two referee scripts were written separately by an independent Claude instance, which re-derived every proof and found no error in the theorems.

- `s6_monodromy_checks.py` (24 checks) verifies:
 - the relations, the invariance of Q_0 and its uniqueness up to scale, and its elementary divisors;
 - Lemma 2.1 symbolically;
 - the generation of $\mathrm{SL}_2(\mathbb{Z})$ by R and S , and $c = h(1, 0, -6)$;
 - the refinements and the coset representatives;
 - the indices of $G \bmod D$ for $D = 2, 3, 4, 6, 8, 9, 12$, by breadth-first enumeration;
 - the level-12 image of Theorem 3.1(4);
 - the orbits of §6 for $D \leq 9$.
- `ver52_referee_claims.py` (22 checks) verifies:
 - that $S \mapsto 9, T \mapsto 1$ defines a homomorphism on all 1152 elements of $\mathrm{SL}_2(\mathbb{Z}/12)$;
 - the decompositions of Theorem 3.1(2);
 - that Φ is multiplicative on random products and vanishes on random words in T_1 and T_2 ;
 - that every one of the 165,888 elements of $G \bmod 12$ has $\Phi = 0$;
 - $H' = \langle c, c' \rangle$;
 - the subgroups N_k ;
 - the non-splitting computation;
 - the dual action;
 - Theorem 6.1 for all $D \leq 14$, as full partitions;
 - Theorem 5.1;
 - Proposition 4.1(4).
- The referee’s scripts confirm Theorem 3.1 by a constructive membership algorithm, and Theorem 6.1 for all $D \leq 24$.

10. Questions

1. Is Φ the restriction to $G_{\mathbb{Z}}$ of the character $\chi_3\chi_4^{-1}$ of Gritsenko and Hulek? It agrees with it on the Heisenberg part (Proposition 4.1(4)). What remains is the relative normalization of $\chi_3\chi_4^{-1}$ at the two positive transvections L_T and t_Y , which lie at cusps of different divisor. If the values agree, $G = \mathrm{Stab}(\gamma) \cap \ker \chi_3\chi_4^{-1}$.
2. Is the first homology of the Arf-one torus of [R, 15_, (147)] the lattice $(\bar{u}, \bar{w})$ of Proposition 4.1? If so, the odd spin structure that cuts out H' is realized geometrically in the construction.
3. Is the $\mathrm{SL}_2(\mathbb{Z})$ -graded piece $\gamma^{\perp}/\langle \gamma \rangle$ the homology of an elliptic family inside or under the tori?
4. The index 12 of G in $G_{\mathbb{Z}}$ equals the order of $\mathrm{SL}_2(\mathbb{Z})^{\mathrm{ab}}$ and of the peripheral character. Theorem 5.1 explains the coincidence at the level of groups. Does it have a meaning in the geometry of the completed threefold, for instance through the mod-24 boundary of [R, 15_, Theorem 6.1]?

References

- [A] A compact complex threefold fibred by tori over the projective line, and the six-sphere, manuscript circulated by L. Alpöge, 2026, <https://alpo.ge/s6.pdf>; reconstructed LaTeX in file 03_ of [R].
- [R] Verification and Reconstruction of a Claimed Complex Structure on S^6 : An AI-Produced Validate/Repair/Disprove Record, Zenodo, DOI 10.5281/zenodo.22678442. Files 14_ (reader’s guide to the topology update) and 15_ (higher torus and cusp update), 5 September 2026.
- [At] M. F. Atiyah, Riemann surfaces and spin structures, *Ann. Sci. École Norm. Sup.* (4) 4 (1971) 47–62.

- [BT] C. Brav, H. Thomas, Thin monodromy in $\mathrm{Sp}(4)$, *Compositio Math.* 150 (2014) 333–343, arXiv:1210.0523.
- [GH] V. Gritsenko, K. Hulek, Commutator coverings of Siegel threefolds, arXiv:alg-geom/9702007.
- [Jo] D. Johnson, Spin structures and quadratic forms on surfaces, *J. London Math. Soc.* (2) 22 (1980) 365–373.
- [Ku] V. S. Kulikov, Degenerations of K3 surfaces and Enriques surfaces, *Math. USSR Izv.* 11 (1977) 957–989.
- [PP] U. Persson, H. Pinkham, Degeneration of surfaces with trivial canonical bundle, *Ann. of Math.* 113 (1981) 45–66.
- [Sa] P. Sarnak, Notes on thin matrix groups, in: *Thin Groups and Superstrong Approximation*, MSRI Publ. 61 (2014).
- [Se] J.-P. Serre, *Trees*, Springer.